BY ORDER OF THE COMMANDER
AIR FORCE MATERIEL COMMAND

AIR FORCE MATERIEL COMMAND
INSTRUCTION 21-103

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Maintenance

RELIABILITY CENTERED MAINTENANCE (RCM) PROGRAMS

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This instruction implements policy provided in DAFPD 21-1, *Maintenance of Military Materiel*. This instruction clarifies Reliability-Centered Maintenance (RCM) program requirements in DAFI 63-101/20-101, *Integrated Lifecycle Management*, by providing guidance and procedures for establishing, implementing, monitoring, and sustaining preventive maintenance (PM) programs for weapon systems and equipment using RCM methodology. This publication applies to all AFMC units of the Regular Air Force. This instruction does not apply to the Air Force Reserve, Air National Guard or the United States Space Force. This AFMCI may NOT be supplemented. Refer recommended changes and questions about this publication to the OPR using the DAF Form 847, *Recommendation for Change of Product*; route DAF Forms 847 from the field through the appropriate functional chain of command. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See DAFMAN90-161, *Publishing Processes and Procedures*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the Publication OPR for non-tiered compliance items. Ensure all records generated as a result of processes prescribed in this publication adhere to AFI 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the DAF.

SUMMARY OF CHANGES

This document has been revised and should be completely reviewed. Changes include updating the document to current publication standards, reference updates, and general cleanup.

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Section A—General RCM Program Information

1. Overview. The purpose of this instruction is to define Reliability-Centered Maintenance (RCM) program responsibilities within Air Force Materiel Command (AFMC). RCM methodology establishes preventive maintenance (PM) requirements to optimize systems/equipment availability and improve reliability and maintainability. This instruction provides requirements, guidance and procedures for establishing, implementing, monitoring, and sustaining PM programs for weapon systems and equipment using RCM. This instruction covers the entire weapon system/equipment life cycle from concept development, design, acquisition, sustainment until system retirement (i.e. cradle to grave). **(T-2)** Terms and abbreviations used in this instruction are listed in **Attachment 1.**

1.1. Scheduled PM requirements must be developed using RCM analysis procedures or similar data-driven analysis. Weapon system programs may obtain and tailor general guidance for implementing RCM programs or performing RCM analysis from NAVAIR 00-25-403, Management Manual, *Naval Aviation Reliability-Centered Maintenance Process*. RCM processes and equivalent or similar data-driven analyses must be compliant with SAE JA1011 *Evaluation Criteria for Reliability-Centered Maintenance (RCM) Processes*. (**Note**: SAE JA1012, *A Guide to the Reliability-Centered Maintenance (RCM) Standard,* provides guidance on using SAE JA1011.) RCM must be employed throughout the weapon system/equipment

life cycle to determine proper balance of PM, and to establish effective failure management strategies. **(T-2)**

1.1.1. NAVAIR 00-25-403 can be obtained from the NATEC Technical Data Website (**https://mynatec.navair.navy.mil/**). A DoD CAC is required to register for an account.

1.1.2. Standard Work Packages (SWP) referenced in NAVAIR 00-25-403 can be obtained from the NAVAIR Sustainment Group site on Flank Speed. A DoD CAC is required to request Guest Access.

1.1.2.1. Request guest access to Flank Speed. While on NIPRNet or VPN, go to: **https://myaccess.microsoft.us/%40flankspeed.onmicrosoft.us#/access-packages/5a163b13-bfd4-43d5-b5ff-4f1966bc7c4e**

1.1.2.2. Submit the request for Guest Access.

1.1.2.3. After Guest Access is granted (automatically within a few minutes) go to: **https://flankspeed.sharepoint-mil.us/sites/NAVAIR_Sustainment_Group_HomePage/Shared%20Documents/Forms/AllItems.aspx**

1.1.2.4. Request access to this page using this justification: "Requesting access to NAVAIR 00-25-403 RCM Standard Work Packages."

1.2. RCM programs for Major Range and Test Facility Bases (MRTFB) as defined by DoDD 3200.11, *Major Range and Test Facility Base (MRTFB)*, are not required but encouraged. MRTFB asset owners and managers should consider the potential return on investment before implementing RCM on an individual basis for each test facility. MRTFB programs may comply with some, all or none of the requirements in NAVAIR 00-25-403 or SAE JA1011.

2. Responsibilities. The Program Manager must:

2.1. Plan, conduct, document, implement, and sustain RCM programs IAW SAE JA1011 *Evaluation Criteria for Reliability-Centered Maintenance Processes* using as guidance NAVAIR 00-25-403..

2.2. Use Failure Mode, Effects and Criticality Analysis (FMECA), RCM process, Product Support Analysis, reliability predictions, maintenance data, failure data, reliability analysis, failure analysis reports, system safety hazard risk analyses, operational studies, engineering analysis and tradeoff studies, and past performance studies to support RCM analyses and programs.

2.3. Use the results of the RCM analysis to determine and reassess PM requirements.

2.4. Summarize results of RCM programs and submits for Air Logistics Complex (ALC) and Major Command (MAJCOM) to evaluate impacts to the PDM or Maintenance/Modification packages in the Engineering Requirements Review Process (ERRP).

2.5. Analyze all proposed changes/refinements to maintenance requirements using data from all available sources including AFTO Forms 22, Enhanced Technical Information Management System (ETIMS) Recommended Changes, Technical Order System Publication Improvement Report and Reply, and interim operational and safety supplements that are a result of mishaps.

2.6. Evaluate and analyze all recommended refinements in the established PM program by using the principles of RCM.

2.7. Maintain RCM documentation.

2.8. Coordinate proposed new maintenance requirements involving engines and time change items with the item manager before publication.

2.9. Coordinate any significant changes in organization workload with the using command, using AFMC organizations, and Air Force Sustainment Center (AFSC) before publishing them in technical order manuals.

2.10. Periodically assess PM requirements, program intervals, and special inspections by conducting reviews and requesting assistance from the item manager, Equipment Specialist (ES), and commodity engineer.

2.11. Initiate periodic joint Program Manager, item manager, ES, commodity engineer, and using command Program Review Conferences or meetings to increase collaboration.

2.12. Obtain all funding needed to plan, develop, execute, and sustain the RCM program.

3. Program Definition. RCM programs ensure that maintenance practices support the safest and most reliable operation of which the weapon system or equipment end item is capable. Through careful application of inspection and scheduled maintenance requirements, critical failures that can be anticipated will be minimized, and the highest probability of war-fighting capability will be achieved. RCM analysis is used to develop scheduled inspection and maintenance requirements. The methodology involves the application of an RCM decision logic process to a problem or failure mode identified by the Failure Modes, Effects and Criticality Analysis (FMECA) for new weapon systems and equipment end items, or a combination of an updated FMECA and field failure data for in-service weapon systems and equipment end items. **(T-2)**

3.1. RCM analysis may be performed on a complete system or systems of a unit, including engines, or on individual items or tasks. Initially, an analysis will be performed on the complete system. In addition to periodic assessments, a subsystem or item analysis will be performed, when dictated, by modifications, maintenance performance data, or other valid indicators. **(T-2)**

3.2. Analysis performed on new weapon systems and equipment end items will rely primarily on predicted failure rates, failure modes and effects and, where feasible, equipment performance from similar weapon systems and equipment end items or testing results. RCM analysis of fielded, in-service systems also relies on FMECA using, equipment performance data, materiel deficiency reporting and the maintenance data collection system. This data and the RCM analysis procedure are used to update existing PM requirements and validate new PM requirements generated by field input, operational experience, or modifications. **(T-2)**

3.3. Reliability growth testing, engineering testing, and developmental test and evaluation (DT&E) data may help to confirm, verify, and update predicted or suspected failure rates, failure modes, failure mechanisms, and time-to-failures. For failure rates and time-to-failure characteristics, rather than assuming an underlying exponential distribution, test data should be analyzed to find an accurate failure distribution. In addition, these tests provide an accurate focus for corrective actions, and PM tasks. DT&E should be used in cases of substandard system performance when information is required in addition to the FMECA to investigate,

system, subsystem, assembly, or item parameters and characteristics and/or determine the quality of modifications or configurations.

3.4. An RCM program covers four main areas: planning, analysis, implementation, and sustainment. NAVAIR 00-25-403 provides ample information on these four areas. **(T-2)**

Section B—RCM Program Requirements

4. Initial Requirements. The organization initiating new developments or modifications must develop the initial inspection and PM requirements based on an RCM analysis or similar data-driven analysis. **(T-2)**

4.1. Periodic Assessment of Requirements. The Program Manager must assess PM requirements for systems and equipment at least every 2 years. Note that for any PM program for operational systems with extensive maintenance histories and structured programs to adjust maintenance tasks and intervals, which were determined with an analysis other than RCM, the reassessment does not have to involve a specific RCM decision logic analysis. Reassessment should include failure and replacement data provided by the using commands. This may be done on a continuous basis if enough documented proof of this procedure is kept. **(T-2)**

4.2. Documentation. The Program Manager is responsible for conducting the RCM analysis and using the results to establish the initial inspection and maintenance requirements. If the system is operational, the Program Manager must use field failure data, operational maintenance data, and RCM analysis and principles to update inspection and PM requirements. The initial RCM analysis and all updating analyses must be documented and archived to preserve the history and rationale for maintenance tasks. This documentation provides a basis from which to monitor the effectiveness of the inspection and maintenance program and to establish an audit trail of all RCM decisions. **(T-2)**

4.3. New acquisition and modification programs must use the RCM process to develop and document PM requirements. **(T-2)**

4.4. Prior to DoDI 5000.02, *Operation of the Adaptive Acquisition Framework*, Milestone C, RCM results must be incorporated into maintenance plans and technical publications. **(T-2)**

4.5. RCM must be sustained and continuously used to update failure management strategies throughout the entire life cycle of the program. Any PM program, which was determined by an analysis process other than NAVAIR 00-25-403 RCM, may be sustained using the original method. **(T-2)**

4.6. RCM analyses must be documented in a digital data format capable of being imported. **(T-2)**

4.7. MIL-STD-1798, Mechanical Equipment and Subsystems Integrity Program (MECSIP), Subtask 4, requires the Program Office (PO) to conduct RCM, Maintenance Steering Group (MSG)-3, Condition Based Maintenance Plus (CBM+) or equivalent analysis to identify PM. **(T-2)**

4.7.1. MSG-3 is published by the Airlines for America (A4A) for original equipment manufacturer (OEM) and commercial airline. MSG-3 is intended for developing an initial projection of scheduled maintenance prior to aircraft introduction into service, therefore it requires a follow-on program for a life-long sustainment. DoDM 4151.25, *Reliability-*

Centered Maintenance, directs the incorporation of RCM throughout the total system life cycle, from requirements development through disposal, and ensures flow down of RCM-enabling requirements including serialized item management, development of FMECA for supportability analysis, inclusion of AI/ML, data analytics, digital twins for RCM analysis, and continuous monitoring of scheduled maintenance compliance and effectiveness. **(T-2)**

4.7.2. Prognostics and Health Management (PHM) and CBM+ programs must be based on RCM analysis, readiness requirements, and life-cycle cost goals. DoDI 4151.22, *Condition-Based Maintenance Plus (CBM+) for Materiel Maintenance*, directs implementation of RCM and establishes RCM as a key enabler of CBM+. **(T-2)**

4.8. AF RCM engine guidance resides in AFMAN 20-116, *Propulsion Life Cycle Management for Aerial Vehicles*.

4.9. MIL-STD-3024, *Propulsion System Integrity Program (PSIP)*, cites AFMCI 21-103.

4.10. AFMAN 63-143, *Centralized Asset Management Procedures*, establishes maintenance requirements based on RCM.

4.11. TO 00-20-1, *Technical Manual, Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures*, establishes that changes to inspections intervals, concepts or requirement must be made by the Single/Program Manager only after thorough analysis of the data from the Maintenance Information System (MIS) and from the appropriate RCM analysis. **(T-2)**

4.12. TO 00-25-4, *Technical Manual, Depot Maintenance of Aerospace Vehicles and Training Equipment*, establishes depot maintenance programs requirements based on RCM. **(T-2)**

4.13. Weapon system program RCM requirements should be established with an agreement and approach for the development and use of a design FMECA and an initial RCM program prior to DoDI 5000.02, Milestone B, and an updated FMECA and RCM program prior to Milestone C to support system safety, reliability and maintainability (R&M), and supportability analyses.

4.14. Scope of RCM Analysis. The analysis program must consist of the following:

4.14.1. A FMECA on significant systems, assemblies, and items. This includes analysis of hardware to find out what failure modes can occur on each item being analyzed and the effect each failure mode has on the item and the total system. When determining the scope of an analysis, extreme care and conservative judgment must be used to ensure that no items with significant impact on safety, environmental compliance, operations, and cost are excluded. **(T-2)**

4.14.2. A decision logic procedure that guides the analyst through a screening process to establish PM requirements based on known or probable failure modes and effects. NAVAIR 00-25-403 contains decision logic, however other similar data-driven RCM processes that meet SAE JA1011 may use alternative decision logic structures. **(T-2)**

4.14.3. A frequency determination to select the best interval for each inspection and PM requirement. **(T-2)**

4.14.4. A periodic assessment of individual inspection requirements and intervals to evaluate basic maintenance concepts (e.g., phase versus periodic), Programmed Depot

Maintenance (PDM), and maintenance tasks program intervals (e.g., 50-hour phase versus 75- or 100-hour phase). Since initial program intervals are based on failure predictions and may not be accurate, an aggressive RCM program to drive accuracy of failure rates based on field data and the identification of the appropriate maintenance strategy and tasks requirements is needed. **(T-2)**

4.14.5. Documentation of RCM analyses and assessments to have consistently traceable maintenance requirements. Previous analyses should be compared to current analyses to establish new maintenance requirements or refine existing ones.

4.14.6. Potential Programmed Depot Maintenance (PDM) and Analytical Condition Inspection (ACI) Tasks. The RCM process may reveal potential PDM or ACI tasks. These tasks must be defined, justified, and submitted for approval by the Engineering Requirements Review Process (ERRP). **(T-2)**

Section C—RCM Program Procedures

5. Initial Analysis. The procedures for accomplishing an initial RCM analysis are as follows:

5.1. Identify the candidate items to be analyzed. All significant items must be subjected to RCM analysis. Significant Items refers to parts whose failure, failure-effect, and failure-consequence could impact safety, environmental compliance, mission performance, operations, or cost. Hardware selection is the determination of which hardware items in the hardware breakdown will be analyzed. If a complete analysis will be performed, hardware selection is simply identifying all the items at the selected level of analysis, minus items that do not warrant analysis. **(T-2)**

5.2. Screen for significant item candidates identified from technical and engineering data and mission equipment lists. When preliminary or published work unit codes are available, each coded item is a candidate. When preliminary or published inspection and maintenance requirements manuals or inspection work cards are available, analyze all items listed as significant to validate, revise, or delete the maintenance requirements. **(T-2)**

5.3. Include the completed lists of significant items as part of the index for the maintenance requirements analysis package for each program. This identifies those items that have been considered and determines where the analysis information is stored for future reference. **(T-2)**

5.4. Failure Mode, Effects and Criticality Analysis. RCM analysis stresses systematic and thorough analysis of significant failure modes and their effects on the safety and reliability of the system. Failure modes, item functions, and proposed inspection tasks must then be subjected to the decision logic tree process to establish the validity of each maintenance requirement. Documentation of the failure modes, effects and criticality analysis, the decision logic process, and the resulting tasks will provide consistent traceable maintenance requirements from which the maintenance program will be developed and refined. When changes are made in system design that may expand, remove, or reduce the impact of identified failure modes, the FMECA must be repeated for the redesigned or modified portions to ensure that all predictable failure modes in the new design or modification are considered. For these modifications, it's important to note that a complete overhaul of the existing FMECA is not always required. FMECA should be treated as a living document updated throughout the

lifecycle as failures are experienced. Updates include newly identified failure modes and updated failure rates on both newly modified and legacy systems. **(T-2)**

5.5. Consider all types of failure modes and effects that pertain to reliability, including but not limited to deterioration and corrosion. Conduct a complete evaluation of each significant assembly or item with primary consideration for safety, operational readiness, mission reliability and performance, environmental impact, and economy. Document failure modes, effects and criticality analysis on items that do not warrant maintenance requirements to preclude future duplication of effort. **(T-2)**

5.6. In conducting the FMECA, overall mission effectiveness and the reliable operation of systems and subsystems must remain paramount. Consider the functional and operational relationships of the significant items and assemblies being analyzed to the overall system. Thus, the analysis should consider the effects of failure of items on higher or lower-level assemblies, systems, or structures. **(T-2)**

5.7. Failure modes, effects, and criticality are a primary design consideration to provide maximum safety and operational effectiveness. They are frequently the determining factor for redundancy of equipment or functions and for safety or protective devices. Any subsequent analysis to determine inspection and maintenance requirements should consider these design considerations because they frequently reduce the impact of failures on safety or mission effectiveness. **(T-2)**

5.8. The analyst applies the RCM logic process, establishes the inspection or time change requirements and the frequency, recommends the equipment maintenance facility (field or depot), and provides documented rationale for each maintenance requirement. The initial analysis program establishes the inspection tasks. The information is kept for future reference on those items analyzed and determined to have no scheduled PM requirements. **(T-2)**

5.9. Failure Analysis and Corrective Action System (FRACAS). The purpose of FRACAS is to establish a closed loop failure reporting system, procedure for analysis of failures to determine cause, and documentation for recording corrective actions taken. FRACAS starts early in the development phase and continues throughout the program's life cycle. Although FRACAS and FMECA are designed and capable of being performed independently of each other, there is a synergistic effect when the two are coupled. FRACAS provides engineering documentation, failure analysis, data for reliability analysis, and a centralized location for deficiencies, failure data, and actions taken by the Government or contractor to correct root causes, prevent recurrence of failure, and/or restore the equipment to an operational status. The FRACAS process promotes reliability improvement throughout the system or equipment life cycle, and corrective actions may drive the need for maintenance requirements updates.

6. In-Service Systems Analysis. After a maintenance program is established, it must be continually evaluated and updated to maintain maximum efficiency and provide minimum impact on operational readiness. This process involves surveillance and resolution of day-to-day problems that impact inspection and maintenance requirements as well as a periodic assessment of the total overall program tasks and intervals. Incoming documentation that may affect RCM programs must be assessed to ensure proper evaluation. **(T-2)**

6.1. New Maintenance Tasks. Each reported problem or modification that may result in establishing new maintenance requirements must be analyzed using RCM. The analysis must

use RCM principles to justify the resulting maintenance requirements. The analysis must be documented as well as the supporting engineering rationale. Any new maintenance requirements or changes to existing maintenance tasks requires product support analyses, like maintenance task analysis (MTA), to determine the product support elements (PSE), such as consumables and tools, needed to accomplish new tasks or implement changes to existing ones. RCM is a required input to other supportability analyses that identify all the requirements, support resources, and support infrastructure needed to perform maintenance and other mitigating and corrective actions. **(T-2)**

6.2. Periodic Assessment. The Program Manager must conduct a periodic assessment (at least every 2 years) of all maintenance requirements to determine if current failure data and experience indicates the need to refine the tasks or intervals. **(T-2)**

6.3. FMECA Updates. During the sustainment phase FMECAs must be continuously updated with new or missed failure modes, causes, and effects information, re-evaluated failure rates assumed during design phase, and new criticality analyses reflective of actual operational performance and real-world service histories. **(T-2)**

7. Decision Logic Process. Coupled with the FMECA on significant items and assemblies, failure data and reports, the RCM decision logic process prescribes the analytical procedures used to validate PM requirements, to determine the appropriate failure management strategy, and to accept, eliminate or lessen the consequences of functional failures. **(T-2)**

7.1. RCM decision logic diagrams and procedures are found in NAVAIR 00-25-403 or similar data-driven RCM processes that meet SAE JA1011. The diagrams contain the essential decision logic requirements for application to RCM analysis. These procedures may be expanded to include additional considerations. Avoid considerations that would compromise safety, mission effectiveness, reliability, or economy. **(T-2)**

7.2. The RCM decision logic diagrams contain a series of questions that are answered with either "yes" or "no" answers. The answers, based on the FMECA or failure data and reports, determine whether an inspection task is required. Although some systems and equipment will not have an initial RCM analysis baseline, all changes to existing inspection and maintenance requirements must be validated by using RCM principles. This analysis ensures a positive approach to establishing and refining maintenance requirements. **(T-2)**

8. Interval Determination. The RCM decision logic process and the FMECA do not consider the frequency of PM. The decision logic process must be supplemented with PM interval analysis to enable an effective RCM program. Since the frequency greatly determines the amount of work expended in a maintenance program, there should be as much emphasis placed on this determination, as on the selection process. The loss of function and any secondary damage caused by a failure must be considered when determining the interval. An initial interval must be established for all new PM and the interval for established PM requires review and analysis for possible refinement. **(T-2)**

9. Contracting for RCM. Sometimes it may be necessary to contract for RCM with the OEM or support contractors. This decision should be carefully considered to ensure the RCM effort is accomplished and sustained to a level that is satisfactory to the Program Manager, equipment operators, and maintainers. When contracting for RCM, the statement of work (SOW) should utilize SAE JA1011; NAVAIR 00-25-403 or similar data-driven processes; MIL-STD-3034, DoD

Standard Practice, *Reliability-Centered Maintenance (RCM) Process*; FMECA; and MIL-STD-882, DoD Standard Practice, *System Safety* documents to ensure the contractor is proposing a process that is compliant with the tenets of RCM. During the contract, the Government activity and personnel responsible for the long-term support and sustainment of the RCM program and the resulting PM program should remain directly involved and provide appropriate expertise in the conduct and review of the contractor analysis efforts. Particularly, there may be a need to ensure appropriate interaction and information is gained from the operators, maintainers, supporting engineers, and logisticians. The deliverables should be scheduled such that appropriate progress is ensured and any problems are identified before investment of resources is put into follow-up activities. Contractual provisions should ensure pertinent RCM data are available to the Government for access and retrieval throughout the system or equipment life cycle.

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Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 33-322, *Records Management and Information Governance Program,* 23 March 2020

AFMAN 20-116, *Propulsion Life Cycle Management for Aerial Vehicles,* 13 April 2022

AFMAN 63-143, *Centralized Asset Management Procedures*, 18 December 2020

DoDI 4151.22, *Condition Based Maintenance Plus (CBM+) for Materiel Maintenance*, 14 August 2020

DoDI 5000.02, *Operation of the Adaptive Acquisition Framework*, 23 January 2020

MIL-STD-1798D, DoD Standard Practice, *Mechanical Equipment and Subsystems Integrity Program*, 1 September 2021

MIL-STD-3024, DoD Standard Practice, *Propulsion System Integrity Program (PSIP)*, 14 May 2018

MIL-STD-3034A, DoD Standard Practice, *Reliability-Centered Maintenance (RCM) Process*, 29 April 2014, (Note: Naval Sea Systems Command publication. For of the purpose of this standard, the term "ship" refers to submarines, surface ships, aircraft carriers, and craft.)

NAVAIR 00-25-403, *Naval Aviation Reliability-Centered Maintenance Process*, 1 September 2023

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DoDI 4151.18, *Maintenance of Military Materiel*, 30 August 2024

DoDM 4151.25, *Reliability-Centered Maintenance (RCM)*, 16 February 2024

MIL-STD-882E, DoD Standard Practice, *System Safety*, 27 September 2023

DoDD 3200.11, *Major Range and Test Facility Base (MRTFB)*, 27 December 2007

Prescribed Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Product*

Abbreviations and Acronyms

ACI—Analytical Condition Inspection

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFMC—Air Force Materiel Command

AFPD—Air Force Policy Directive

ALC—Air Logistics Complex

ATA—Air Transport Association of America

CA—Criticality Analysis

CBM+—Condition Based Maintenance Plus

DAF—Department of the Air Force

DAFMAN—Department of the Air Force Manual

DAFPD—Department of the Air Force Policy Directive

DoDD—Department of Defense Directive

DoDI—Department of Defense Instruction

DoDM—Department of Defense Manual

DT&E—Developmental Test and Evaluation

ELMP—Engine Life Management Plan

ERRP—Engineering Requirements Review Process

ES—Equipment Specialist

ETIMS—Enhanced Technical Information Management System

FMEA—Failure Modes, Effects, and Analysis

FMECA—Failure Modes, Effects, and Criticality Analysis

FRACAS—Failure Reporting, Analysis and Corrective Action System

IAW—In Accordance With

MAJCOM—Major Command

MDS—Mission, Design, and Series

MECSIP—Mechanical Equipment and Subsystems Integrity Program

MRTFB—Major Range and Test Facility Base

MSG—Maintenance Steering Group

MTA—Maintenance Task Analysis

NAVAIR—Naval Air Systems Command

NAVSEA—Naval Sea Systems Command

NIPRNet—Non-secure Internet Protocol Router Network

OEM—Original Equipment Manufacturer

OPR—Office of Primary Responsibility

PDM—Programmed Depot Maintenance

PHM—Prognostics and Health Management

PM—Preventive Maintenance

PO—Program Office

PSE—Product Support Elements

PSIP—Propulsion System Integrity Program

R&M—Reliability and Maintainability

RCM—Reliability-Centered Maintenance

RDS—Records Disposition Schedule

RDT&E—Research Development Test and Evaluation

SOW—Statement of Work

TO—Technical Order

VPN—Virtual Private Network

Terms

Aerospace Equipment (Weapon Systems and Equipment)—The term refers to weapon systems and equipment such as aerospace vehicles, equipment, missiles, nuclear weapons, Test Measurement and Diagnostic Equipment (TMDE), ground communication electronics (C-E), trainers, training equipment, engines, Flight Support Equipment (FSE), industrial plant equipment, and all related Support Equipment (SE). (TO 00-20-1).

Aerospace Vehicle—Any vehicle that is designed to operate in the atmosphere and/or space environment.

Aircrew Flight Equipment (AFE)—Individual items worn by, attached to, used by or provided for aircrew and passengers to maintain life, health, function, and safety during flight and to provide for escape, descent, survival, and recovery. Also identified as FSE.

Analytical Condition Inspection (ACI)—The systematic disassembly and inspection of a representative sample of aircraft to find hidden defects, deteriorating conditions, corrosion, fatigue,

overstress and other deficiencies in the aircraft structure or systems. ACIs are normally over and above those inspections specified in the technical order or PDM work specifications.

Best Practice—Procedures that are accepted or prescribed as most effective.

Common Support Equipment (CSE)—Aerospace maintenance support equipment that is used across multiple weapons systems such as Aerospace Ground Equipment (AGE); Test, Measurement and Diagnostic Equipment (TMDE), Automated Test Systems (ATS); Propulsion Support Equipment (PSE); and Munition Materiel Handling Equipment (MMHE).

Criticality Analysis—A procedure that prioritizes each failure mode identified in the FMEA according to the combined influence of its severity and its probability of occurrence.

Depot Industrial Plan Equipment (DIPE)—Depot Industrial Plant Equipment is known as equipment utilized in depot industrial maintenance area as outlined in AFMCI 21-127, Depot Maintenance Plant Management.

Failure Cause—The physical or chemical processes, design defects, quality defects, part misapplication, or other processes which are the basic reason for failure or which initiate the failure mechanisms by which deterioration proceeds to failure.

Failure Consequences—The impact of a functional failure (including secondary damage) caused by failure mode(s) based on evidence of failure and adverse effect on safety, mission effectiveness, environment, operations, and economics.

Failure Effects—The result of a functional failure on surrounding items, the functional capability of the end item, and hazards to personnel and the environment.

Failure Mechanism—Failure mechanisms are the chemical, electrical, physical, mechanical, structural, or thermal processes leading to failure.

Failure Mode, Effects and Analysis (FMEA)—A process used to determine the function(s) of each item, the functional failures associated with each function, the failure modes that have the potential to cause each functional failure, and the effect and severity of each failure mode. This analysis identifies all potential, predicted and actual functional failures of an item.

Failure Mode, Effects and Criticality Analysis (FMECA)—A process which combines a Failure Mode and Effects Analysis (FMEA) and a Criticality Analysis (CA).

Failure Mode—The manner by which a failure is observed.

Failure Rate—The frequency with which an item or component fails, expressed in failures per unit of time.

Flank Speed—The Department of the Navy's Microsoft 365 cloud tenant for collaboration, including Teams and SharePoint. See Guest Access information at: **https://usaf.dps.mil/sites/TipsToolsApps/SitePages/GuestAccess.aspx**

Functional Failure—The inability of an item to perform a specific function within specified limits.

Function—An intended purpose of an item as described by a required standard of performance.

Maintenance Steering Group (MSG-3)—Air Transport Association of America (ATA) Publication MSG-3, Operator/Manufactured Schedule Maintenance Development.

Materiel—All items necessary to equip, operate, maintain, and support military activities without distinction as to its application for administrative or combat purposes.

Peculiar Support Equipment (PSE)—PSE is any aerospace maintenance support equipment designed to service and/or support a specific MDS/Weapon System.

Preventive Maintenance (PM)—Actions performed prior to functional failure (multiple failures or demand requirements for hidden failures) to achieve the desired level of safety and reliability for an item.

Prognostics and Health Management (PHM) Systems—Diagnostic or prognostic devices and systems that are used to monitor equipment condition and provide indications to the operator or maintainer. These systems may also initiate automatic actions to deal with the condition(s) sensed or predicted.

Program Manager—The Program Manager, as defined in DoDD 5000.01, is the designated individual with responsibility for and authority to accomplish program objectives for development, production, and sustainment to meet the user's operational needs. The Program Manager has responsibility and authority to accomplish objectives for the total life cycle of the program. The Program Manager is responsible for assuring the OSS&E of systems, subsystems and end items.

Test, Measurement and Diagnostic Equipment (TMDE)—TMDE are those devices used to maintain, evaluate, measure, calibrate, test, inspect, diagnose or otherwise examine materials, supplies, equipment, and systems to identify or isolate actual or potential malfunction, or decide if they meet operational specifications established in technical documents.

Trainer—Equipment designed and procured specifically for formal training programs.

Weapon System—A combination of one or more weapons with all related equipment, materials, services, personnel, and means of delivery and deployment (if applicable) required for self-sufficiency. (Joint Pub 1-02).